

## Linear algebra and geometry 1 - TTPU

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**Marks** -150.00/1000.00

**Grade** -4.95 out of 33.00 (-15%)

### Question 1

Incorrect

Mark -15.00 out of  
100.00

Fixed a coordinate system in the space, consider the points of coordinates  $P_0 = (0, 0, 0)$ ,  $P_1 = (2, 0, 1)$  e  $P_2 = (1, 1, 1)$ . Then, the area of the triangle of vertices  $P_0$ ,  $P_1$  and  $P_2$  is equal to

- ☐ (a)  $\frac{\sqrt{6}}{2}$ .
- ☒ (b)  $\sqrt{6}$ . ✖
- ☐ (c)  $2\sqrt{3}$ .
- ☐ (d) 6.

Risposta errata.

The correct answer is:  $\frac{\sqrt{6}}{2}$ .

**Question 2**

Incorrect

Mark -15.00 out of  
100.00

Consider the following line  $r : x - y = x + y - z = 0$ .

Find the correct statement.

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- ☐ (a)  $r$  lies on the plane  $z = 0$ .
  - ☒ (b)  $r$  passes through the point  $(1, 1, 0)$ . ✖
  - ☐ (c)  $r$  lies on the plane  $2x + z = 0$ .
  - ☐ (d)  $r$  has the same direction of the vector  $(1, 1, 2)$ .

Risposta errata.

The correct answer is:  $r$  has the same direction of the vector  $(1, 1, 2)$ .

**Question 3**

Incorrect

Mark -15.00 out of  
100.00

Consider the following linear system in the variables  $x, y, z$ :

$$\begin{cases} x - y + z = 3 \\ x + z = 4 \\ 2x - 3y + 2z = 5 \end{cases}$$

Find the right statement.

- 
- ☐ (a) The system has one unique solution.
  - ☒ (b) The system has only two solutions. ✖
  - ☐ (c) The system has infinitely many solutions.
  - ☐ (d) The system has no solutions.

Risposta errata.

The correct answer is: The system has infinitely many solutions.

**Question 4**

Incorrect

Mark -15.00 out of 100.00

Let us take the matrix  $A = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -1 & -1 \end{pmatrix}$ ; find the right statement:

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- ☐ (a)  $A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & -1 & -1 \end{pmatrix}$ .
- ☐ (b)  $A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$ .
- ☒ (c)  $A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix}$ . ✖
- ☐ (d)  $A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & -1 & -1 \end{pmatrix}$ .

Risposta errata.

The correct answer is:  $A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & -1 & -1 \end{pmatrix}$ .

**Question 5**

Incorrect

Mark -15.00 out of 100.00

Consider a linear mapping  $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ . Find which one of the following statements cannot hold.

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- ☐ (a)  $f$  is injective.
- ☐ (b) The matrix associated to  $f$  has rank 1.
- ☒ (c)  $\dim \operatorname{Im}(f) = 2$ . ✖
- ☐ (d)  $\dim \ker(f) \geq 1$ .

Risposta errata.

The correct answer is:  $f$  is injective.

**Question 6**

Incorrect

Mark -15.00 out of  
100.00

Let us consider the vectors  $\vec{v}_1 = (1, 1, 1, 0)$ ,  $\vec{v}_2 = (2, 2, 0, 2)$ ,  $\vec{v}_3 = (3, 0, 3, 3)$  e  $\vec{v}_4 = (0, 4, 4, 4)$  in  $\mathbb{R}^4$ . Find the right statement.

- ☐ (a)  $\{\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4\}$  is a basis of  $\mathbb{R}^4$ .
- ☐ (b) The vector  $\vec{v}_1$  lies in the subspace generated by  $\{\vec{v}_2, \vec{v}_3, \vec{v}_4\}$ .
- ☐ (c)  $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$  is a basis of  $\mathbb{R}^2$ .
- ☒ (d) There exist  $a, b, c \in \mathbb{R}$  such that  $\vec{v}_4 = a\vec{v}_1 + b\vec{v}_2 + c\vec{v}_3$ . ✖

Risposta errata.

The correct answer is:  $\{\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4\}$  is a basis of  $\mathbb{R}^4$ .**Question 7**

Incorrect

Mark -15.00 out of  
100.00

Let  $q(x, y) = x^2 - 8xy - y^2$  be a quadratic form. Find the correct statement.

- ☐ (a) The matrix associated to  $q$  is  $\begin{pmatrix} -1 & -4 \\ -4 & 1 \end{pmatrix}$ .
- ☒ (b)  $q$  is negative definite. ✖
- ☐ (c)  $q$  is positive definite.
- ☐ (d)  $q$  is indefinite.

Risposta errata.

The correct answer is:  $q$  is indefinite.**Question 8**

Incorrect

Mark -15.00 out of  
100.00

Let us consider the conic  $\mathcal{C}$  in the Euclidean plane defined by the equation  $2x^2 + 2xy + 2x + 2y^2 - 1 = 0$ .

Find the correct statement.

- ☐ (a)  $\mathcal{C}$  is a parabola.
- ☒ (b)  $\mathcal{C}$  is degenerate. ✖
- ☐ (c)  $\mathcal{C}$  is a hyperbola.
- ☐ (d) None of the three other statements is correct.

Risposta errata.

The correct answer is: None of the three other statements is correct.

**Question 9**

Incorrect

Mark -15.00 out of  
100.00Let  $f : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$  such that $f(x, y, z) = (2x + 2y + 2z, 2x + 2y + 2z, 2x + 2y + 2z)$ . Let  $\lambda_i$  be an eigenvalue of  $f$  and let  $E(\lambda_i)$  be the corresponding eigenspace.

Find the correct statement.

- ☐ (a)  $\lambda_1 = 6, \lambda_2 = 2, \lambda_3 = 0, \dim E(\lambda_1) = 2, \dim E(\lambda_2) = 1, \dim E(\lambda_3) = 1$
- ☐ (b)  $\lambda_1 = 2, \lambda_2 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2$
- ☒ (c)  $\lambda_1 = 6, \lambda_2 = 2, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2$  ✖
- ☐ (d)  $\lambda_1 = 6, \lambda_2 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2$
- ☐ (e)  $\lambda_1 = 6, \lambda_2 = 2, \lambda_3 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 1, \dim E(\lambda_3) = 1$
- ☐ (f)  $\lambda_1 = 6, \lambda_2 = 2, \lambda_3 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2, \dim E(\lambda_3) = 1$

Risposta errata.

The correct answer is:  $\lambda_1 = 6, \lambda_2 = 0, \dim E(\lambda_1) = 1, \dim E(\lambda_2) = 2$

**Question 10**

Incorrect

Mark -15.00 out of  
100.00

For  $a, b \in \mathbb{R}$  consider the following linear system in the variables  $x, y, z$ :

$$\begin{cases} x - ay + z = 0 \\ x + z = 0 \\ bx - y + 2z = 0 \end{cases}$$

and let  $S_{a,b}$  be the set of solutions of the linear system of equations.

Find the right statement.

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- ☐ (a) For  $b = 3$  and  $a = 1$  the linear system admits infinitely many solutions and  $\dim S_{a,b} = 2$ .
  - ☐ (b) For  $b = 3$  and  $a = 2$  the linear system does not have solutions.
  - ☒ (c) For  $b = 3$  and  $a = 2$  the linear system admits infinitely many solutions and  $\dim S_{a,b} = 3$ . ✖
  - ☐ (d) For  $b = 1$  and  $a = 0$  the linear system admits infinitely many solutions and  $\dim S_{a,b} = 2$ .
  - ☐ (e) For  $b = 3$  and  $a = 1$  the linear system admits infinitely many solutions and  $\dim S_{a,b} = 1$ .
  - ☐ (f) For  $b = 1$  and  $a = 0$  the linear system admits infinitely many solutions and  $\dim S_{a,b} = 1$ .

Risposta errata.

The correct answer is: For  $b = 1$  and  $a = 0$  the linear system admits infinitely many solutions and  $\dim S_{a,b} = 1$ .